ENGS 3302 PATTERN RECOGNITION · WEEK GUIDE

Week 2: Features, Distance and the Nearest-Mean Classifier

Week of 15 October 2026 · Videos ≈ 86 min · Lab video ≈ 45 min

This week in one sentence. Turn objects into numbers, measure how far apart they are, put the features on one scale, and classify a new sample by its nearest class centre.

Learning outcomes: T2 · L1 · L2 (see the syllabus).

What to do, in this order

	Step	Notes	Time
☐	`W02_Math_Toolkit.pdf`, then the **Week 2 Math check** on Moodle (no grade)	Toolkit	25 min
☐	**Video 2.1** Features: turning objects into numbers	§2.1	15 min
☐	**Video 2.2** Measuring distance	§2.2	12 min
☐	**Video 2.3** When the units decide (uses the interactive page *Distance and Scaling Explorer*)	§2.3	15 min
☐	**Video 2.4** Normalization	§2.4	17 min
☐	**Video 2.5** The nearest-mean classifier; how good is it?	§2.5–2.7	27 min
☐	Read the Notes again, and do *Check yourself* at the end	all	40 min
☐	**Lab video** and `Lab_W02.ipynb` (no grade)	—	≈ 2 h
☐	`Assignment_W02.ipynb` **(graded, about 1%) submit by Wed 21 Oct, 23:59**	—	≈ 1.5 h

How to submit the assignment

1. In Colab: *Runtime* → *Restart and run all*, so every cell shows its output.
2. *File* → *Download* → *Download .ipynb*.
3. Upload the file to *Assignment 2* on Moodle and press *Submit*.

Marking: 1 = runs, every task attempted, every *explain in your own words* answered · 0.5 = more than half attempted · 0 = not submitted, empty or copied. Late work: up to 48 hours without penalty. In the in-person sessions you may be asked to explain your work.

Files and help

Files: `W02_Notes.pdf` (**the study material; the exam covers what is in the Notes**) · `W02_Math_Toolkit.pdf` · slides (for viewing only) · `Lab_W02.ipynb` · `Assignment_W02.ipynb` · `penguins.csv` (data for both notebooks; in Colab they download it themselves) · interactive page *Distance and Scaling Explorer* (download and open in your browser; works offline).

Help: ask on the course Discord server or by e-mail. **In-person session this week: male section, Thursday 15 October, 12:00–14:00** (its first session). The female section meets next on Thursday 22 October.